01 / PROBLEM AND CONTRIBUTION

ATmega328P data-logger PCB

Independent hardware design · KiCad | PCB design study

DESIGN AT A GLANCE

Integrated an ATmega328P, two external EEPROM devices, and a real-time clock into a four-copper-layer KiCad board, with Gerber exports available in the repository.

Engineering question

How can a data-logger board bring processing, persistent storage, timekeeping, and accessible interfaces together?

Situation

The board design provides the hardware foundation for a standalone data logger. The checked-in schematic identifies an ATmega328P-AU, two 24LC1025 devices, a DS1337S clock, and dedicated clock sources.

Task

Develop the schematic and PCB layout for the processor, external memory, clock, and connector interfaces.

My contribution

Schematic capture, PCB implementation, and design documentation in the linked personal GitHub project.

Tools and concepts

KiCad / PCB layout / ATmega328P / I²C interfaces / Gerber outputs

02 / IMPLEMENTATION

Circuit to board

01 / Design step

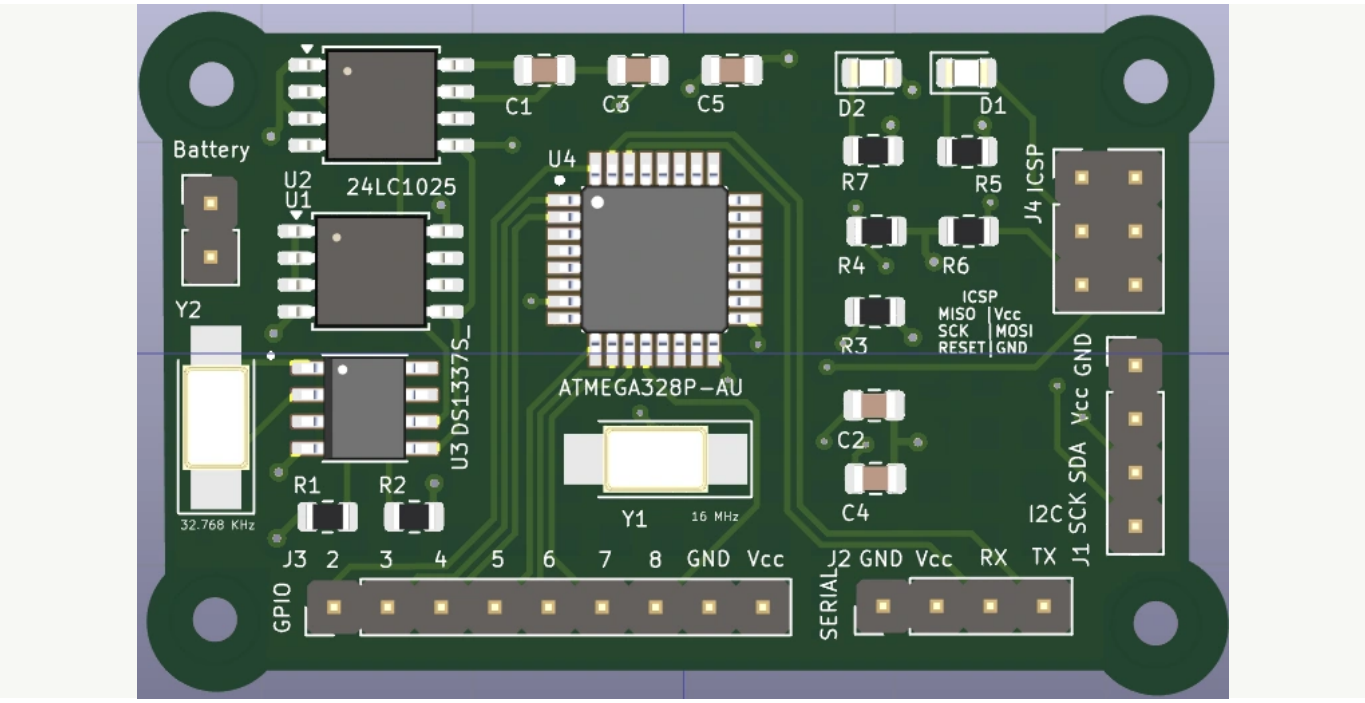
Organized the controller around a 16 MHz clock and the RTC around a 32.768 kHz crystal. Included memory devices, pull-up resistors, decoupling, and connector sheets in the KiCad project.

02 / Design step

Placed and routed the design in the checked-in four-copper-layer board. The source contains 29 footprints, including mounting features, with 274 track segments and 48 vias.

03 / Design step

Included two-layer and four-layer Gerber export directories alongside a component list. The current editable board uses four copper layers; the older exports are retained as separate artifacts.



Data-logger board render

Repository front-side 3D render showing the controller, memory, clock, and connectors.

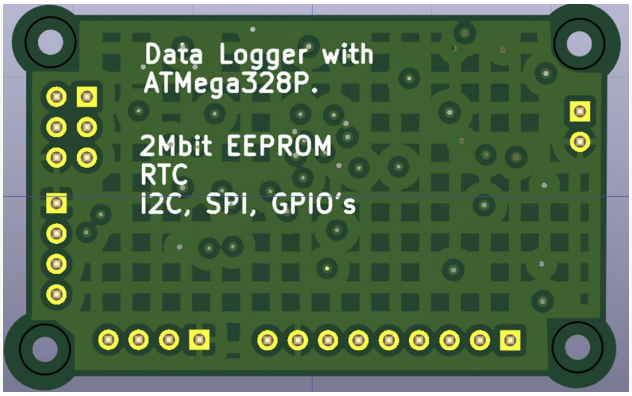
03 / EVIDENCE AND SCOPE

Design evidence and next steps

Current board	4 copper layers
Controller	ATmega328P-AU
External memory devices	2 × 24LC1025

Available evidence

The repository provides editable schematic and board files, front/back renders, a component list, and Gerber exports. The design files establish a concrete hardware implementation, while operational data logging still requires firmware and bench evidence.



Data-logger reverse-side render

Repository reverse-side 3D render; this is a design visualization, not a fabricated-board photograph.

Scope and limitations

The README's 512 KB capacity and two-layer description do not match the current schematic and board. The portfolio uses the device part numbers and current layer count instead. Sampling rate, endurance, logging firmware, and successful hardware operation are not demonstrated by the reviewed files.

Next steps - proposed work

- Reconcile the memory-capacity and board-revision descriptions in the README.
- Add firmware, a documented write/read test, and timestamp validation.

Source: PCB-design-projects GitHub repository: MCU-Datalogger-With-Built-In-512K-EEPROM; README, KiCad schematic/board, and repository previews. Reviewed 11 November 2026.

Open the project repository